

Proving That a Periodic Orbit Exists

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Definition: Positive semi-orbit

Let $z(t; z_0)$ be a solution defined for every $t \geq 0$ and satisfying $z(0; z_0) = z_0$. Its **positive semi-orbit** is

$$\gamma^+(z_0) = \{z(t; z_0) : t \geq 0\}.$$

Definition: Omega-limit set

Let $z(t; z_0)$ be a solution defined for every $t \geq 0$ and satisfying $z(0; z_0) = z_0$. Its **omega-limit set** is

$$\omega(z_0) = \left\{ p \in \mathbb{R}^2 : \begin{array}{l} \text{there is a sequence } t_n \rightarrow \infty \\ \text{such that } z(t_n; z_0) \rightarrow p \end{array} \right\}.$$

Proposition: Basic omega-limit properties

Let $U \subseteq \mathbb{R}^2$ be open, let $F : U \rightarrow \mathbb{R}^2$ be locally Lipschitz, and let $z(t; z_0)$ solve $\dot{z} = F(z)$. If $\gamma^+(z_0)$ lies in a compact subset of U , then $\omega(z_0)$ is nonempty, compact, and invariant.

Theorem: Poincaré–Bendixson theorem

Let $U \subseteq \mathbb{R}^2$ be open and let $F \in C^1(U)$. Suppose the positive semi-orbit of z_0 is contained in a compact subset of U . If $\omega(z_0)$ contains no equilibrium of

$$\dot{z} = F(z),$$

then $\omega(z_0)$ is a periodic orbit.

Recall: Positive invariance

A set D is **positively invariant** for $\dot{z} = F(z)$ if every solution satisfies

$$z(0) \in D \implies z(t) \in D \quad \text{for every } t \geq 0 \text{ for which the solution is defined.}$$

Definition: Trapping region

For a flow on an open set $U \subseteq \mathbb{R}^2$, a **trapping region** is a nonempty, compact, connected, positively invariant set $D \subset U$.

Observation: Inward-pointing boundary test

Let D be a closed planar region bounded by finitely many smooth arcs meeting at corners, and let F be locally Lipschitz on a neighborhood of D . At a smooth point of ∂D , let n be the outward unit normal. The strict inequality

$$F \cdot n < 0$$

means that the field crosses into D . To allow tangency, require $F \cdot n \leq 0$ at every smooth boundary point. At a corner, require this inequality for the outward normal of every incident smooth side. These conditions prevent a solution from leaving D through its boundary.

Corollary: Trapping-region criterion

Let $F \in C^1(U)$ for an open set $U \subseteq \mathbb{R}^2$, and let $D \subset U$ be a trapping region. If D contains no equilibrium of $\dot{z} = F(z)$, then D contains at least one periodic orbit.

Proof. Choose $z_0 \in D$. Positive invariance keeps the solution in the compact set $D \subset U$. The continuation theorem therefore makes the solution global forward in time, and

$$\omega(z_0) \subseteq D.$$

The set $\omega(z_0)$ contains no equilibrium, so the Poincaré–Bendixson theorem makes it a periodic orbit in D . $\square$

Visualization: Boundary directions for an annular trap

For

$$D = \{(r, \theta) \in (0, \infty) \times S^1 : r_- \leq r \leq r_+\}, \quad 0 < r_- < r_+,$$

the outward normals and inward conditions are

boundary	outward normal	inward condition
$r = r_+$	e_r	$\dot{r} < 0$
$r = r_-$	$-e_r$	$\dot{r} > 0$

Here e_r denotes the radial unit vector pointing toward increasing r . On the inner boundary, “outward from the annulus” means “into the hole.”

Question: Construct a trapping annulus

Consider

$$\dot{r} = r(1 - r^2) + \mu r \cos \theta, \quad \dot{\theta} = 1, \quad \theta \in S^1, \quad 0 < \mu < 1.$$

Prove that the system has a periodic orbit.

Solution. Choose the radii

$$r_- = \sqrt{\frac{1-\mu}{2}}, \quad r_+ = \sqrt{2(1+\mu)}.$$

On $r = r_-$, the bound $\cos \theta \geq -1$ gives

$$\begin{aligned} \dot{r} &= r_-(1 - r_-^2 + \mu \cos \theta) \\ &\geq r_-(1 - r_-^2 - \mu) \\ &= \frac{r_-}{2}(1 - \mu) > 0. \end{aligned}$$

On $r = r_+$, the bound $\cos \theta \leq 1$ gives

$$\begin{aligned} \dot{r} &= r_+(1 - r_+^2 + \mu \cos \theta) \\ &\leq r_+(1 - r_+^2 + \mu) \\ &= -r_+(1 + \mu) < 0. \end{aligned}$$

Thus

$$D = \{(r, \theta) : r_- \leq r \leq r_+, \theta \in S^1\}$$

is a trapping annulus. The map

$$(r, \theta) \mapsto (r \cos \theta, r \sin \theta)$$

identifies $(0, \infty) \times S^1$ with the punctured plane, so D is a planar annulus and the planar trapping-region criterion applies. Since $\dot{\theta} = 1$, the system has no equilibrium in D . The trapping-region criterion gives at least one periodic orbit in D . It gives no uniqueness or stability conclusion.

Question: Construct a polygonal trap for the glycolysis model

For

$$\dot{x} = -x + ay + x^2y, \quad \dot{y} = b - ay - x^2y, \quad a, b > 0,$$

construct a compact positively invariant polygon in the closed nonnegative quadrant $\{(x, y) : x \geq 0, y \geq 0\}$.

Solution. Choose

$$Y > \frac{b}{a}$$

and let D have vertices

$$(0, 0), \quad (b + Y, 0), \quad (b, Y), \quad (0, Y).$$

Question: Construct a polygonal trap for the glycolysis model (continued)

Its four sides satisfy:

side	calculation	direction
$x = 0$	$\dot{x} = ay \geq 0$	right or tangent
$y = 0$	$\dot{y} = b > 0$	up
$y = Y$	$\dot{y} = b - (a + x^2)Y \leq b - aY < 0$	down
$x + y = b + Y$	$\dot{x} + \dot{y} = b - x \leq 0$	toward $x + y \leq b + Y$

On the diagonal, $x \geq b$. At its top endpoint $x = b$, the diagonal derivative is zero but the top-side inequality is strict. Thus the inward inequality holds for every side incident to each boundary point, so D is positively invariant.

Recall: Hyperbolic source

An equilibrium of a planar system is a **hyperbolic source** if both eigenvalues of its Jacobian have positive real parts.

Question: When does the glycolysis equilibrium repel?

For

$$\dot{x} = -x + ay + x^2y, \quad \dot{y} = b - ay - x^2y, \quad a, b > 0,$$

find the equilibrium and determine when it is a hyperbolic source.

Solution. Let

$$F(x, y) = (-x + ay + x^2y, b - ay - x^2y).$$

Adding the equations gives

$$\dot{x} + \dot{y} = b - x.$$

Thus every equilibrium satisfies $x_* = b$, and substitution gives the unique equilibrium

$$z_* = (x_*, y_*) = \left(b, \frac{b}{a + b^2}\right).$$

The Jacobian is

$$DF(x, y) = \begin{pmatrix} -1 + 2xy & a + x^2 \\ -2xy & -(a + x^2) \end{pmatrix}.$$

At z_* , the determinant and trace of the Jacobian are

$$\Delta = \det DF(z_*) = a + b^2 > 0$$

and

$$\tau = \operatorname{tr} DF(z_*) = -\frac{b^4 + (2a - 1)b^2 + a + a^2}{a + b^2}.$$

Let $q = b^2$. The condition $\tau > 0$ is

$$q^2 + (2a - 1)q + a + a^2 < 0.$$

The roots of the quadratic are

$$q_{\pm} = \frac{1 - 2a \pm \sqrt{1 - 8a}}{2}.$$

Question: When does the glycolysis equilibrium repel? (continued)

Therefore z_* is a hyperbolic source when

$$0 < a < \frac{1}{8}, \quad q_- < b^2 < q_+.$$

Definition: Strict exit neighborhood

Let $U \subseteq \mathbb{R}^2$ be open, let $F : U \rightarrow \mathbb{R}^2$ be continuous, and let $z_* \in U$ be an equilibrium of $\dot{z} = F(z)$. A compact neighborhood $E \subset U$ of z_* with smooth boundary is a **strict exit neighborhood** if

$$F(z) \cdot n_E(z) > 0 \quad (z \in \partial E),$$

where n_E is the outward unit normal of E .

Lemma: Lyapunov matrix for a repelling linear system

Let A be a real 2×2 matrix whose eigenvalues have positive real parts. Define

$$P = \int_0^\infty e^{-A^\top s} e^{-As} ds.$$

The integral converges to a symmetric positive definite matrix satisfying

$$A^\top P + PA = I.$$

Here I is the 2×2 identity matrix.

Proof. The eigenvalues of $-A$ have negative real parts, so e^{-As} decays exponentially and the integral converges. For every $\xi \neq 0$,

$$\xi^\top P \xi = \int_0^\infty \|e^{-As} \xi\|^2 ds > 0.$$

Moreover,

$$\begin{aligned} A^\top P + PA &= - \int_0^\infty \frac{d}{ds} (e^{-A^\top s} e^{-As}) ds \\ &= I. \end{aligned}$$

□

Proposition: Quadratic exit neighborhood of a planar hyperbolic source

Let $U \subseteq \mathbb{R}^2$ be open, let $F \in C^2(U)$, and let $z_* \in U$ be an equilibrium of $\dot{z} = F(z)$. Suppose every eigenvalue of

$$A = DF(z_*)$$

has positive real part. Define

$$P = \int_0^\infty e^{-A^\top s} e^{-As} ds$$

and

$$V(z) = (z - z_*)^\top P (z - z_*).$$

Then $\dot{V} > 0$ sufficiently near z_* , except at z_* itself. Consequently, for every sufficiently small $\rho > 0$, the ellipsoid $\{z : V(z) \leq \rho\}$ is a strict exit neighborhood.

Proof. Set

$$\xi = z - z_*.$$

Taylor's theorem gives

$$F(z_* + \xi) = A\xi + R(\xi), \quad R(\xi) = O(\|\xi\|^2) \quad (\xi \rightarrow 0).$$

Using $A^\top P + PA = I$, we obtain

$$\begin{aligned} \dot{V} &= 2\xi^\top P F(z_* + \xi) \\ &= \xi^\top (A^\top P + PA)\xi + 2\xi^\top P R(\xi) \\ &= \|\xi\|^2 + O(\|\xi\|^3) \quad (\xi \rightarrow 0). \end{aligned}$$

For sufficiently small nonzero ξ , the remainder has absolute value at most $\frac{1}{2}\|\xi\|^2$. Hence

$$\dot{V} \geq \frac{1}{2}\|\xi\|^2 > 0.$$

Because P is positive definite, sufficiently small ellipsoids $V \leq \rho$ lie in this neighborhood and in U . On their boundaries, $\nabla V \neq 0$ and the outward unit normal is

$$n = \frac{\nabla V}{\|\nabla V\|},$$

so

$$F \cdot n = \frac{\dot{V}}{\|\nabla V\|} > 0.$$

Hence $V \leq \rho$ is a strict exit neighborhood. □

Question: Force a periodic orbit in the glycolysis model

Assume

$$\dot{x} = -x + ay + x^2y, \quad \dot{y} = b - ay - x^2y,$$

where

$$0 < a < \frac{1}{8}, \quad b > 0, \quad q_- < b^2 < q_+, \quad q_{\pm} = \frac{1 - 2a \pm \sqrt{1 - 8a}}{2}.$$

Let $Y > b/a$, and let D be the positively invariant polygon with vertices

$$(0, 0), \quad (b + Y, 0), \quad (b, Y), \quad (0, Y).$$

Prove that the system has a periodic orbit.

Solution. Let

$$F(x, y) = (-x + ay + x^2y, b - ay - x^2y).$$

The unique equilibrium is

$$z_* = \left(b, \frac{b}{a + b^2}\right),$$

and it is a hyperbolic source.

Because

$$0 < \frac{b}{a + b^2} < \frac{b}{a} < Y,$$

the point z_* satisfies all four defining inequalities for D strictly: $x_* > 0$, $y_* > 0$, $y_* < Y$, and $x_* + y_* < b + Y$. Hence $z_* \in \text{int}(D)$.

By the hyperbolic-source proposition, choose a strict exit ellipsoid E about z_* lying inside $\text{int}(D)$. Remove its interior:

$$K = D \setminus \text{int}(E).$$

The outer boundary still points inward or tangent. On the inner boundary, the outward normals satisfy $n_K = -n_E$, so

$$F \cdot n_K = -F \cdot n_E < 0.$$

Thus K is positively invariant. Since D and E are convex and E lies strictly inside D , K is a compact connected annulus. Its only possible equilibrium was removed with E . The trapping-region criterion therefore gives a periodic orbit in K .

Question: Check the standard glycolysis parameters

For

$$\dot{x} = -x + ay + x^2y, \quad \dot{y} = b - ay - x^2y,$$

let

$$a = 0.08, \quad b = 0.6.$$

Classify the equilibrium and state the conclusion justified by an annular trapping region.

Solution. Let

$$F(x, y) = (-x + ay + x^2y, b - ay - x^2y).$$

The unique equilibrium is

$$z_* = \left(b, \frac{b}{a + b^2}\right) = \left(0.6, \frac{15}{11}\right).$$

Question: Check the standard glycolysis parameters (continued)

The determinant of its Jacobian is

$$\Delta = \det DF(z_*) = a + b^2 = 0.08 + 0.36 = 0.44.$$

Also,

$$\begin{aligned} b^4 + (2a - 1)b^2 + a + a^2 &= 0.1296 - 0.3024 + 0.08 + 0.0064 \\ &= -0.0864, \end{aligned}$$

so

$$\tau = \operatorname{tr} DF(z_*) = \frac{0.0864}{0.44} \approx 0.1964 > 0.$$

Finally,

$$\tau^2 - 4\Delta < 0.$$

The equilibrium is an unstable spiral. The annular trapping region proves that at least one periodic orbit exists. It does not prove that the orbit is unique, isolated, or stable.

Observation: Planar restriction

The Poincaré–Bendixson conclusion is special to planar flows; it does not extend to autonomous systems in dimensions three and higher.

Visualization: What the standard planar criteria establish

criteria	conclusion
$\dot{V} < 0$ on every non-equilibrium state	no periodic orbit
$\nabla \cdot (BF)$ has one strict sign on a simply connected domain	no periodic orbit in that domain
compact trap with no equilibrium	at least one periodic orbit
Poincaré–Bendixson alone	no conclusion about uniqueness or stability